

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech (Electronics & Communication) Sem - III University Examination May,11

EC 204 Electronics Instruments & Measurements

Date: 20.05.11, Friday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 70

- Instructions:** (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION -I

- Q-1 Answer the following 10**
- (a) Define the terms Accuracy, Standard deviation.
 - (b) State two types of systematic errors giving example of each.
 - (c) What is error? Explain different types of static error.
 - (d) What is transducer? List the difference types of transducers.
 - (e) Give the comparison between LED and LCD
- Q-2(a) Draw and explain block diagram of digital Storage Oscilloscope. 6**
(b) Explain frequency synthesized signal generator in detail. 6
- OR**
- Q-2(a) Explain Function Generator with block diagram 6**
(b) Why a delay line used in the vertical section of Oscilloscope? Draw and explain delay line circuit. 6
- Q-3(a) Describe the operation of Wheatstone bridge and derive balance equation for resistance measurement. 5**
- (b) Find the series equivalent inductance and resistance of the network that causes an opposite angle (Hay bridge) to null with the following bridge arms. (Refer Figure A) 5**
 $\omega=3000 \text{ rad/s}$, $R_2=10 \text{ k}\Omega$, $R_1=2 \text{ k}\Omega$, $C_1=1\mu\text{F}$, $R_3=1 \text{ k}\Omega$
- (c) Which are advantages of Thermistor 3**
- OR**
- Q-3(a) Explain wien's bridge with figure and derive equation for frequency measurement 5**
- (b) What is limitation of Maxwell bridge? Maxwell bridge is used to measure inductive impedance. The bridge constants at balance are: $C_1=0.01\mu\text{F}$, $R_1=470\text{k}\Omega$, $R_2=5.1\text{k}\Omega$, $R_3=100\text{k}\Omega$. Find the series equivalent of unknown impedance. 5**
- (c) What conditions must be satisfied to make an ac bridge balance? 3**

SECTION-II

- Q-4(a) Which are the digital display methods? Explain BCD to seven segment converter 8
- (b) Explain digital frequency meter with circuit 5
- OR**
- Q-4(a) Explain construction and operation of LVDT with neat figure 8
- (b) Explain the principle of operation of digital time measurement with block diagram. 5
- Q-5(a) Explain thermocouple in detail 5
- (b) Derive an expression for the gage factor in case of strain gage. 5
- OR**
- Q-5(a) Explain BCD to dot matrix converter in detail. 5
- (b) Draw the block diagram of Data Acquisition System and explain it in detail 5
- Q-6 Write a short note on (Any Three) 12
- Classification of Transducers.
 - Frequency selective Wave analyzer
 - Dual Trace CRO
 - Schering bridge
 - Piezo-electric transducers

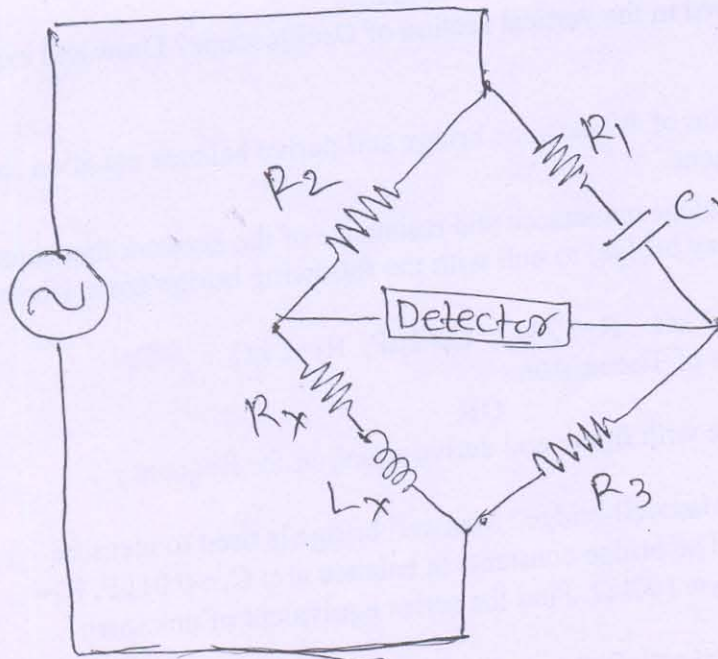


Figure # A